

Graphs as Adjacency Processes

A Point Process Framework for Local Limits of Random Graphs

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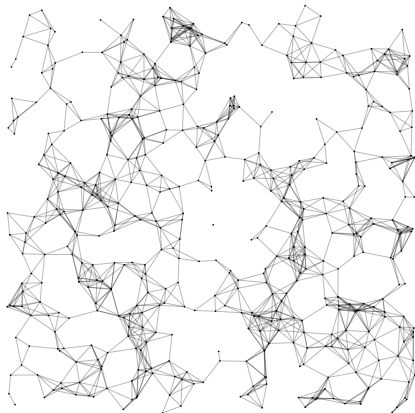
University of Amsterdam (UvA)

Based on bachelor thesis work at TU/e;
Upcoming joint work with Pim van der Hoorn

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Background - Spatial Random Graphs in $\mathbb{R}^d$

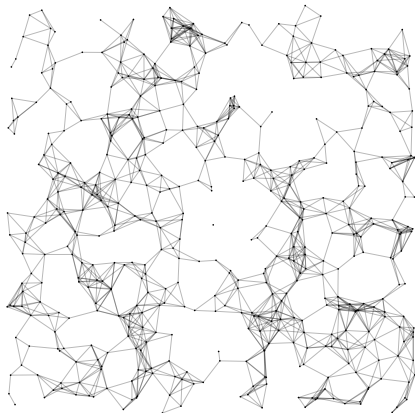
Random Geometric Graph



$$\mathbb{P}(E_{x,y}) = \mathbf{1}\{\|x - y\| \leq \rho\}$$

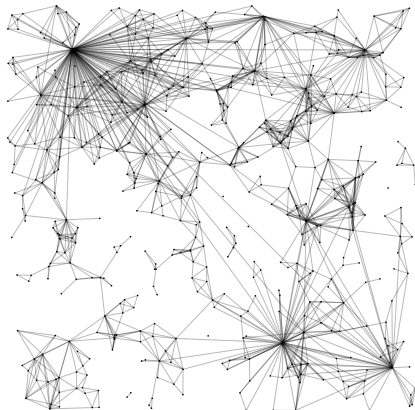
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Random Geometric Graph



$$\mathbb{P}(E_{x,y}) = \mathbf{1}\{\|x - y\| \leq \rho\}$$

SIRG/GIRG

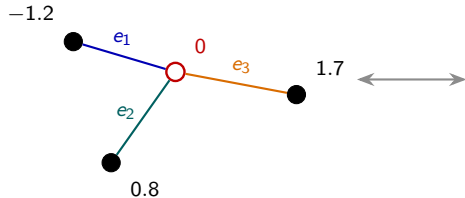


$$\mathbb{P}(E_{x,y}) = \kappa(\|x - y\|, w_x, w_y)$$

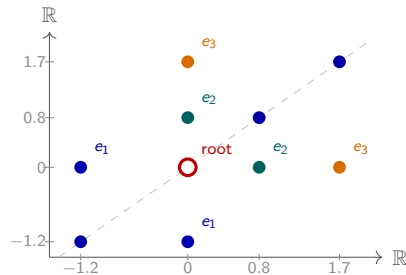
Motivation

- ▶ *Local weak convergence* asks to find (G_∞, p_∞) for which $(G_n, U_n) \Rightarrow (G_\infty, p_\infty)$
- ▶ A graph G on vertices in $\mathbb{R}^d$ can be represented as a point process in $\mathbb{R}^{2d}$

Rooted Labelled Graph (G, p)



Symmetric Adjacency Measure (ξ, p)



- ▶ Palm theory gives a notion of "typical point" for point processes. Can this be associated with a uniform graph root for local weak convergence?

The Main Theorem

Consider a sequence of finite stationary adjacency processes $(\xi_n)_{n \in \mathbb{N}}$, placed along a nice* compact exhaustion of a proper metric space $\mathbb{X}^2$. Suppose that

1. $\xi_n^{(o,o)} \Rightarrow \xi_\infty$ (ξ_∞ will then also have a point at (o, o) almost surely);
2. The vertices concentrate via

$$\frac{|V(\xi_n)|}{\mathbb{E}|V(\xi_n)|} \xrightarrow{L^1} 1;$$

3. $(\xi_n^{(o,o)}, o)_{n \in \mathbb{N}}$ is *radially tight** (vertices reachable from o contained in a compact subset of $\mathbb{X}$ with high probability).

Then, the corresponding graphs converge locally weakly,

$$G(\xi_n, U_n) \Rightarrow G(\xi_\infty, o),$$

and ξ_∞ acts as a nice, sampleable representation of the local weak limit.

The Model-Facing Corollary

Consider a sequence of finite stationary *vertex* processes $(\Gamma_n)_{n \in \mathbb{N}}$, placed along a nice* compact exhaustion of a proper metric space $\mathbb{X}$, and let C be a nice*, continuous-ish, isometric connection rule (measurable SIRG kernel). Suppose that

1. $\Gamma_n^o \Rightarrow \Gamma_\infty$;

2. The vertices concentrate via

$$\frac{|\Gamma_n|}{\mathbb{E}|\Gamma_n|} \xrightarrow{L^1} 1;$$

3. $(\xi_n, o)_{n \in \mathbb{N}} \sim C(\Gamma_n^o, \cdot)$ is *radially tight** (usually suffices that $\mathbb{P}(E_{xy}) \propto t^{-\alpha}$ against $t = d_{\mathbb{X}}(x, y)$).

Then, the corresponding graphs converge locally weakly,

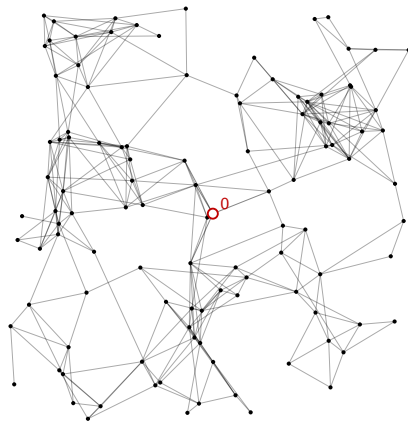
$$G(\xi_n, U_n) \Rightarrow G(\xi_\infty, o),$$

where $\xi_\infty \sim C(\Gamma_\infty, \cdot)$.

The Classical Case - Binomial in $\mathbb{R}^d$

- ▶ $\Gamma_n = \sum_{i=1}^n \delta_{U_i}$ with $U_i \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(\mathbb{T}_n^d)$
- ▶ C is a SIRG rule with $\mathbb{P}(E_{xy}) \propto t^{-\alpha}$, where $\alpha > d$

$$G(\xi_n, U_n) \Rightarrow G(\xi_\infty, 0)$$



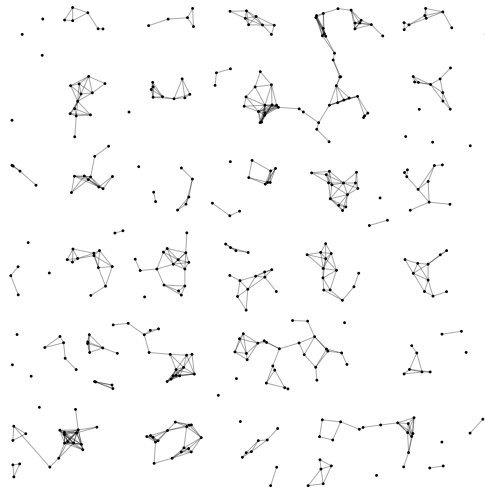
Poisson SIRG limit

$$\Gamma_\infty^0 \stackrel{d}{=} \text{PPP}(1) + \delta_0$$

Novel Results - More Vertex Processes

Sticking with $\mathbb{R}^d$ and $\mathbb{T}_n^d$ for now, we can handle:

- ▶ Mixed-Binomial vertex processes
 $\sum_{n=0}^N \delta_{U_i}$
- ▶ Cox vertex processes (Phase-Shifted, Shot-Noise, Lévy)
- ▶ Neyman-Scott / Clustering

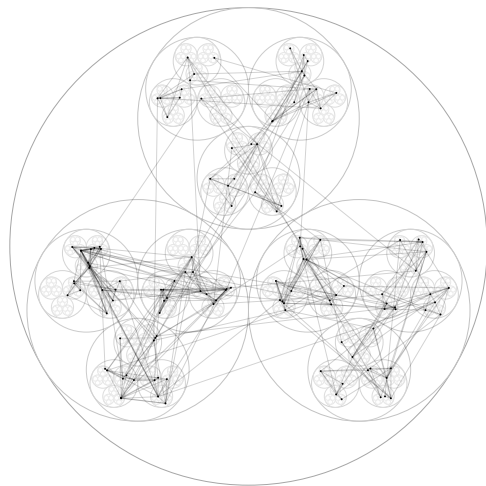


Soft RGG with Cox vertices

Novel Results - Non-Euclidian Spaces

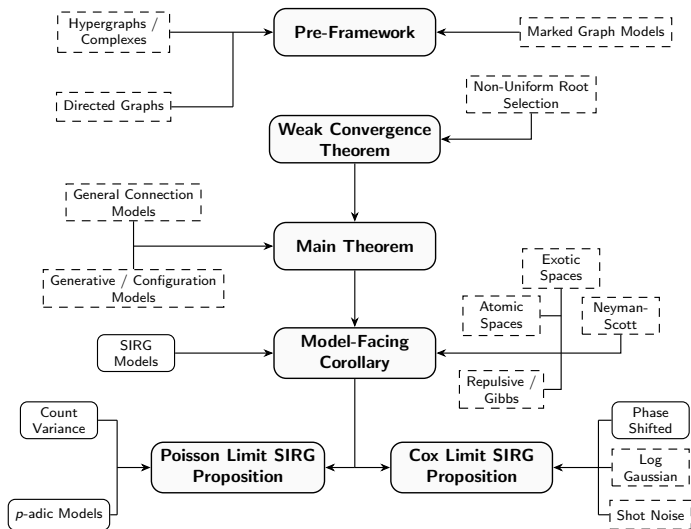
We can also handle the following approximating regimes:

- ▶ $B_{\mathbb{Q}}(0, p^n) \rightarrow \mathbb{Q}_p^d$
- ▶ Proper metric spaces that admit homogeneous quotient spaces
- ▶ $\mathbb{H}^d$? Not yet... no homogeneous quotient spaces



SIRG in $\mathbb{Q}_3$

How YOU Can Benefit 🍷



Overview of the main convergence results and model families feeding into them. Dashed borders indicate topics not directly explored in this paper.

Next Steps

- ▶ Preprint/Paper in the works with Pim van der Hoorn
- ▶ Directed graphs will be included in this revised framework
- ▶ Hyperbolic SIRGs, eventually...
- ▶ Formalize conditions for more vertex processes (Cox, Neyman-Scott, ...)

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